CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

Case 21-2023: A 61-Year-Old Man with Eyelid Swelling

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PRESENTATION OF CASE

Dr. Jonathan E. Lu (Ophthalmology, Massachusetts Eye and Ear): A 61-year-old man was evaluated in an ophthalmology clinic affiliated with this hospital because of eyelid swelling.

Eight years before the current presentation, the patient noticed mild puffiness of the eyelids that was worst in the morning and typically resolved by midday. Five years before the current presentation, the eyelid swelling began to progressively worsen and to last throughout the day. The patient had a history of glaucoma; when the eyelid swelling began to interfere with his vision, he asked for a consultation with the ophthalmologist who had treated his glaucoma. A diagnosis of dermatochalasis with fat prolapse was suspected, and the patient was referred to the ophthalmic plastic and reconstructive surgery clinic affiliated with this hospital for consideration of blepharoplasty.

In the clinic, the patient reported that eyelid puffiness was interfering with his ability to read and drive. He did not have dry eye. He had a sensation of pressure behind the eyes and intermittent double vision. There was generalized joint stiffness in the mornings that resolved after 1 hour. There was left lateral hip pain that radiated to the left buttock and thigh, with associated paresthesia in the left foot.

Human immunodeficiency virus (HIV) infection had been diagnosed 31 years before the current presentation and was well controlled with antiretroviral therapy, although lipodystrophy had occurred with treatment. Other medical history included carpal tunnel syndrome, cataracts, depression, dyslipidemia, hypertension, allergic rhinitis, obstructive sleep apnea, and asthma, in addition to primary open-angle glaucoma. Surgical history included carpal tunnel release surgery (performed 2 months before the current presentation), intraocular lens replacement in both eyes, and lipectomy for the reduction of dorsocervical lipohypertrophy due to lipodystrophy. Medications included antiretroviral therapy (elvitegravir, cobicistat, emtricitabine, and tenofovir alafenamide), bupropion, fluoxetine, triamterene, amlodipine, atorvastatin, inhaled albuterol, and topical testosterone. The administration of iodinated contrast material had caused hives. The patient lived part

Figure 1. Clinical Photographs at Presentation.

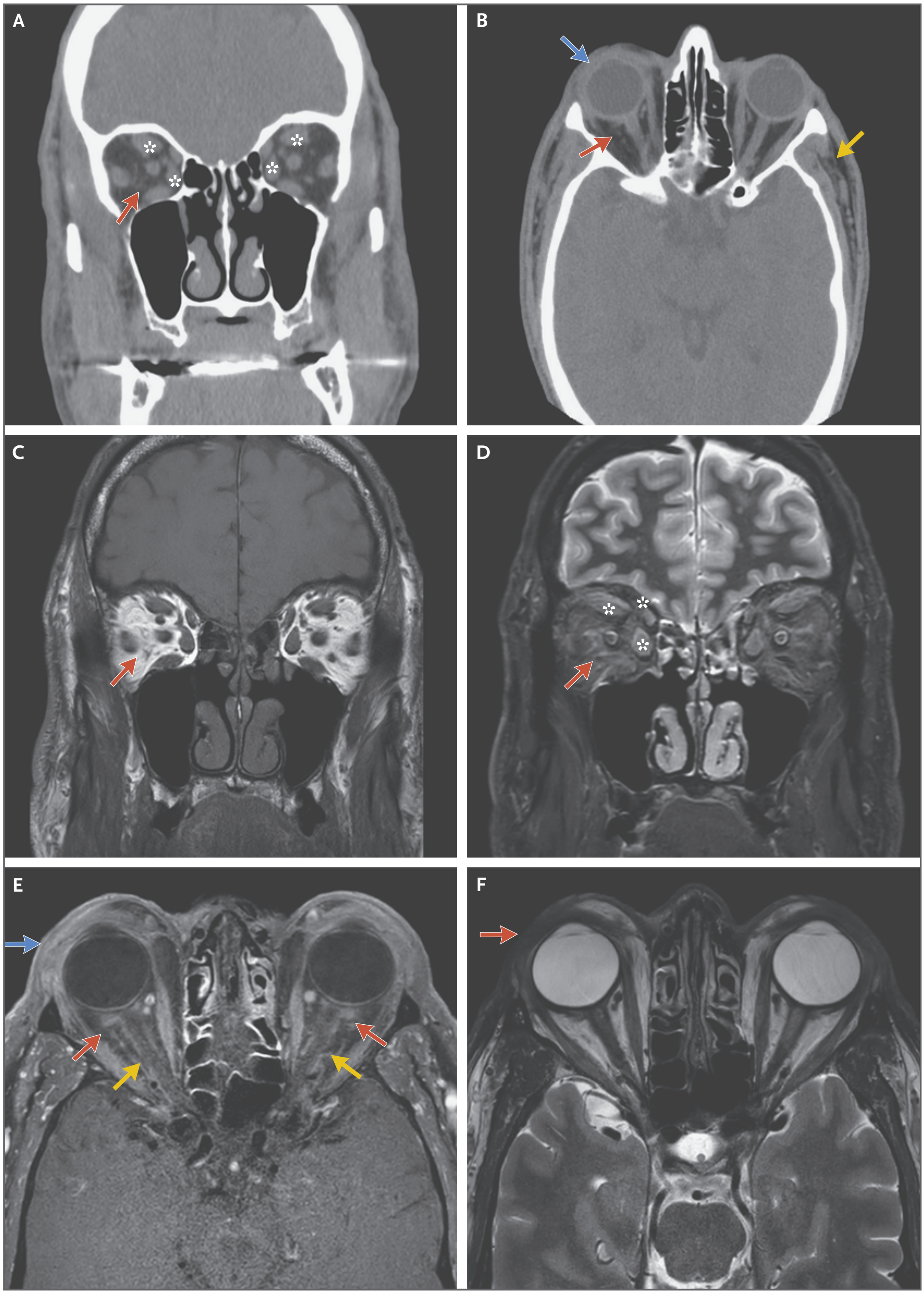
Photographs of the eyes were obtained in several gaze positions. In primary gaze, the patient has edema and mild erythema of the upper eyelids, along with festoons of the lower eyelids (Panel A). The upper-eyelid edema is more pronounced when viewed from below (Panel B). There is mild limitation of extraocular motility in all gaze positions, including up gaze (Panel C), right gaze (Panel D), left gaze (Panel E), and down gaze (Panel F).

time in Florida and part time in New England. He was a lifelong nonsmoker, drank alcohol occasionally, and did not use illicit substances.

On examination, the corrected visual acuity was 20/25 in the right eye and 20/30 in the left eye. The pupils were symmetric and reactive to light. Confrontation testing revealed deficits in the outer superior temporal visual fields and the superior nasal visual fields, which abated with manual elevation of the eyelids. Supraduction and abduction were limited in both eyes. Soft edema of the upper and lower eyelids was present, along with mild erythema of the upper eyelids (Fig. 1). The upper-eyelid skin was overhanging the lashes medially, a finding indicative of dermatochalasis. There was upper-eyelid ptosis. The globes were moderately resistant to retropulsion, and there was mild proptosis. The levator palpebrae superioris muscles had impaired function, with an eyelid elevation of 8 mm in the right eye and 11 mm in the left eye (reference range, 12 to 18). Slit-lamp examination revealed 1+ chemosis and trace injection in the conjunctivae; there was no evidence of superficial punctate keratopathy or cells in the anterior chambers. Funduscopic examination revealed an increased cup-to-disk ratio in the optic nerves, a finding consistent with the patient's known history of glaucoma. The tongue was enlarged, and there was dorsocervical lipohypertrophy on the posterior aspect of the neck.

The blood levels of thyrotropin, free thyroxine, triiodothyronine, thyroid peroxidase antibodies, and thyroid-stimulating immunoglobulins were normal. Imaging studies were obtained.

Dr. Laura V. Romo: Computed tomography (CT) of the face and orbits (Fig. 2A and 2B), performed without the administration of intravenous contrast material, revealed bilateral enlargement of the extraocular muscles and fat stranding in the orbits and elsewhere in the face. Magnetic resonance imaging (MRI) of the



face and orbits (Fig. 2C through 2F), performed before and after the administration of intravenous contrast material, revealed orbital fat stranding and mild, symmetric, diffuse extraocular-muscle enlargement. A short-tau inversion recovery sequence showed an increased signal that spared

Figure 2 (facing page). Imaging Studies of the Face and Orbits.

CT of the face and orbits was performed. A coronal image (Panel A) shows abnormal, curvilinear stranding in otherwise normal, low-attenuation orbital fat (red arrow). Mild, diffuse enlargement of the extraocular muscles, including the superior and medial rectus muscles in both eyes, is also present (asterisks). An axial image (Panel B) shows abnormal, curvilinear stranding in otherwise normal, low-attenuation orbital fat (red arrow); preseptal soft-tissue thickening along the anterior margin of both globes (blue arrow); and fat stranding along both temples (yellow arrow). MRI of the face and orbits was also performed. A coronal T1-weighted image (Panel C) shows abnormal stranding (gray) in normal orbital fat (white), which has high signal intensity (red arrow). A coronal short-tau inversion recovery image (Panel D) shows abnormal stranding (white), which has high signal intensity that spares the myotendinous junctions, in normal orbital fat (gray), where the signal is suppressed (red arrow). Diffuse enlargement of the extraocular muscles, including the superior and medial rectus muscles, is also present and has high signal intensity (asterisks). An axial T1-weighted, contrast-enhanced, fat-saturated image (Panel E) shows abnormal stranding and enhancement in the retrobulbar and episcleral fat (red arrows), diffuse thickening and enhancement of the preseptal soft tissue (blue arrow), and enhancement along the optic-nerve sheaths (yellow arrows), a finding consistent with perineuritis. An axial T2-weighted image (Panel F) shows diffuse preseptal soft-tissue thickening with low signal intensity along the anterior margin of both globes (red arrow).

the myotendinous junctions. Diffuse thickening and stranding of the skin on the preseptal eyelids and face, episcleritis along the posterior margin of both globes, and perineuritis around the optic nerves were also noted.

Dr. Lu: The blood levels of glucose and electrolytes and the results of kidney-function tests were normal, as were the blood levels of angiotensin-converting enzyme (ACE), lysozyme, rheumatoid factor, and C-reactive protein. The erythrocyte sedimentation rate was 22 mm per hour (reference range, 0 to 20). The blood IgG level was 1980 mg per deciliter (reference range, 767 to 1590), the IgG1 level 283 mg per deciliter (reference range, 341 to 894), and the IgG2 level 1750 mg per deciliter (reference range, 171 to 632); the blood IgG3 and IgG4 levels were normal. Indirect immunofluorescence testing for antineutrophil cytoplasmic antibodies (ANCAs) was negative, including an enzyme-linked immunosorbent assay for antibodies to proteinase 3 and myeloperoxidase. A rapid plasma reagin test was nonreactive. An interferon-γ release assay for *Mycobacterium tuberculosis* complex infection was negative.

A diagnostic procedure was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Natalie Wolkow: I participated in the care of this patient, and I am aware of the final diagnosis. I evaluated this patient for upper-eyelid blepharoplasty for the treatment of dermatochalasis (excess upper-eyelid skin); however, the clinical examination findings were atypical, prompting further investigation. He had dermatochalasis, but the skin was puffy, edematous, and mildly erythematous. He also had lower-eyelid edema (festoons) and mild conjunctival chemosis and injection, findings suggestive of inflammation. These features, in combination with upper-eyelid ptosis, mild limitation of extraocular motility in several gaze positions, impairment of levator-muscle function, and moderate resistance to retropulsion when pressing over the eyes, were suggestive of a chronic orbital inflammatory process.

THYROID EYE DISEASE

The most common cause of orbital inflammation is thyroid eye disease. The patient's eyelid edema and erythema, orbital resistance to retropulsion, limited extraocular motility, impaired levator-muscle function, and conjunctival chemosis and injection were consistent with thyroid eye disease. There was no eyelid retraction, lag, or flare, which are typical findings of thyroid eye disease. Instead, there was eyelid ptosis. Patients with thyroid eye disease occasionally present with ptosis, but it is uncommon.[1] The patient had no history of systemic thyroid disease, but more than 20% of patients present with thyroid eye disease before systemic disease is diagnosed.[2]

It is notable that the patient's clinical course was atypical for thyroid eye disease. Most patients with thyroid eye disease have progression of inflammation for 1 to 2 years followed by improvement, whereas this patient had slow progression for many years. In addition, the results of tests for thyroid function and thyroid antibodies were normal in this case. On review of the patient's imaging studies, the bilateral extraocular-muscle enlargement with fusiform features was typical of thyroid eye disease. However, the uniform pattern of extraocular-muscle enlargement and the presence of episcleritis, perineuritis, orbital fat stranding, and skin thickening were atypical and prompted consideration of alternative causes of orbital inflammation.

CANCER

Metastatic cancer, lymphomatous or leukemic infiltration, or paraneoplastic disease can cause extraocular-muscle enlargement, at times with an inflammatory clinical appearance.[3-6] Patients with these diseases may present with unilateral or bilateral limitation of extraocular motility, conjunctival chemosis and injection, resistance to retropulsion, or eyelid edema. Metastatic lesions that affect the extraocular muscles typically manifest as nodular or focal enlargement; in rare cases, lymphoma or leukemia manifests as bilateral, diffuse extraocular-muscle involvement. He had no history of cancer, but extraocular-muscle enlargement can be the presenting sign of cancer.

However, this patient's clinical course was atypical for cancer or paraneoplastic disease. Metastatic cancer or lymphomatous or leukemic infiltration of the extraocular muscles would progress over a period of weeks to months, and when diffuse bilateral disease is present, there can be rapid symptom progression and vision loss.[4] By contrast, this patient had an 8-year history of progressive symptoms without the development of systemic signs of cancer.

CHRONIC INFLAMMATORY ORBITAL MYOSITIS

Several systemic inflammatory diseases can manifest with orbital inflammation, including extraocular-muscle enlargement due to chronic orbital myositis. Sarcoidosis can cause bilateral extraocular-muscle enlargement, but in such cases, it would usually also cause lacrimal gland enlargement or orbital masses.[7,8] The extraocular-muscle enlargement can progress slowly, at times with no pain. In this patient, the normal ACE and lysozyme levels and the absence of lacrimal gland enlargement and orbital masses make sarcoidosis an unlikely diagnosis.

IgG4-related disease is another inflammatory process that can have orbital involvement with slow progression and minimal pain. Orbital IgG4-related disease most often results in lacrimal gland enlargement and tends to cause enlargement of trigeminal nerve branches, particularly the infraorbital nerve, but extraocular-muscle enlargement can also occur.[9] Although a normal IgG4 level alone would not rule out IgG4-related disease, the combination of the normal IgG4 level and the absence of lacrimal gland and infraorbital nerve enlargement lowers the likelihood of IgG4-related disease in this case.

Orbital granulomatosis with polyangiitis usually causes lacrimal gland enlargement or orbital masses, but isolated extraocular-muscle enlargement can occur occasionally.[10] In this patient, the negative ANCA testing, the absence of lacrimal gland enlargement and orbital masses, and the absence of symptoms and imaging findings of sinus disease make extraocular-muscle enlargement due to granulomatosis with polyangiitis unlikely.

MEDICATION-INDUCED ORBITAL MYOSITIS

Medications such as bisphosphonates[11,12] and immune checkpoint inhibitors[13] can cause orbital myositis, although it is typically acute and painful. None of this patient's medications have known associations with orbital inflammation. The initiation of antiretroviral therapy in patients with HIV infection has been associated with immune reconstitution inflammatory syndrome (IRIS). Cases of orbital myositis due to delayed-onset IRIS that developed 1 and 3 years after the initiation of antiretroviral therapy have been described,[14,15] but patients with this condition usually present with acute eye pain, erythema, and edema. IRIS is unlikely in this patient, given the absence of acute symptoms and the long-standing use of antiretroviral therapy.

HIV-RELATED LIPODYSTROPHY

Lipodystrophy can manifest as lipoatrophy, lipohypertrophy, or both.[16] The mechanisms of lipodystrophy are not fully understood, but inflammatory responses to HIV infection and effects of antiretroviral therapy in adipose tissue are thought to lead to these changes.[16] Certain antiretroviral therapies are associated with lipohypertrophy and others with lipoatrophy. This patient had a history of marked dorsocervical lipohypertrophy, for which he had undergone lipectomy. At the time of the current presentation, abnormal reticulation and thickening in the subcutaneous tissue of the face and neck and in the orbital fat were observed on CT and MRI. I considered whether these changes were consistent with HIV-related lipohypertrophy. The face is not typically a site of lipohypertrophy. Orbital lipoatrophy has been reported,[17,18] but orbital lipohypertrophy has not.

Overall, none of the diagnoses that I initially considered fit well with this patient's presentation. His systemic symptoms were revisited, and he was referred to the rheumatology clinic affiliated with this hospital for evaluation of joint stiffness, hip pain, and foot paresthesia. Before his evaluation in the rheumatology clinic, he was evaluated for new dyspnea on exertion and received a diagnosis of cardiomyopathy. Given the presence of new cardiomyopathy, paresthesia, and carpal tunnel syndrome, amyloidosis was added to the differential diagnosis.

AMYLOIDOSIS

Amyloidosis can involve any tissue of the ocular adnexa, but such involvement is very uncommon. Patients usually present to ophthalmologists with localized disease, most often with unilateral eyelid ptosis or an eyelid mass.[19] Less than 10% of patients with ocular adnexal amyloidosis have systemic amyloidosis, with involvement of other organs.[20,21]

Although amyloidosis can in rare cases cause extraocular-muscle enlargement, ocular adnexal amyloidosis is not typically considered to be an inflammatory process. This patient's ocular findings were suggestive of inflammation, so I had not initially considered the diagnosis. Some patients with amyloidosis affecting the extraocular muscles can present with eyelid edema,[22] and when there is extensive amyloid deposition in the ocular adnexa, the eyelids can take on an inflammatory appearance, with festoons and red discoloration of the skin, findings that were seen in this case.[20]

The bilateral extraocular-muscle enlargement, the subcutaneous and orbital fat stranding, and the inflammatory appearance of the eyelids in this patient could be explained by amyloid deposition in the extraocular muscles, orbital fat, and skin, respectively. In addition, the glaucoma could possibly be the result of amyloid deposition in the arterial circle of Zinn–Haller, which supplies the optic-nerve head. The conjunctival chemosis and injection, as well as the episcleritis and perineuritis seen on MRI, could be the result of amyloid deposition in the optic-nerve sheath (dura), sclera, and episclera.[23]

Only 2 to 16% of patients with ocular adnexal amyloidosis have extraocular-muscle involvement.[20,21] Among such patients, those with bilateral, diffuse extraocular-muscle enlargement are more likely to have systemic amyloidosis than those with unilateral extraocular-muscle enlargement.[22] This patient had bilateral orbital disease, no discrete orbital or eyelid masses, and no visible amyloid deposits on eyelid eversion, features that made unilateral localized amyloidosis unlikely (although it is more common than systemic disease) and thus were suggestive of systemic amyloidosis. His slow disease progression, history of carpal tunnel syndrome, and new cardiomyopathy were also consistent with systemic amyloidosis.

Systemic amyloidosis was the leading diagnosis for this patient. To establish the diagnosis of systemic amyloidosis, amyloid must be detected in tissue, so biopsy of affected ophthalmic tissue was indicated. Because biopsy of extraocular muscle requires general anesthesia and carries a risk of permanent diplopia, biopsy of eyelid skin and orbital fat was performed with the patient under local anesthesia.

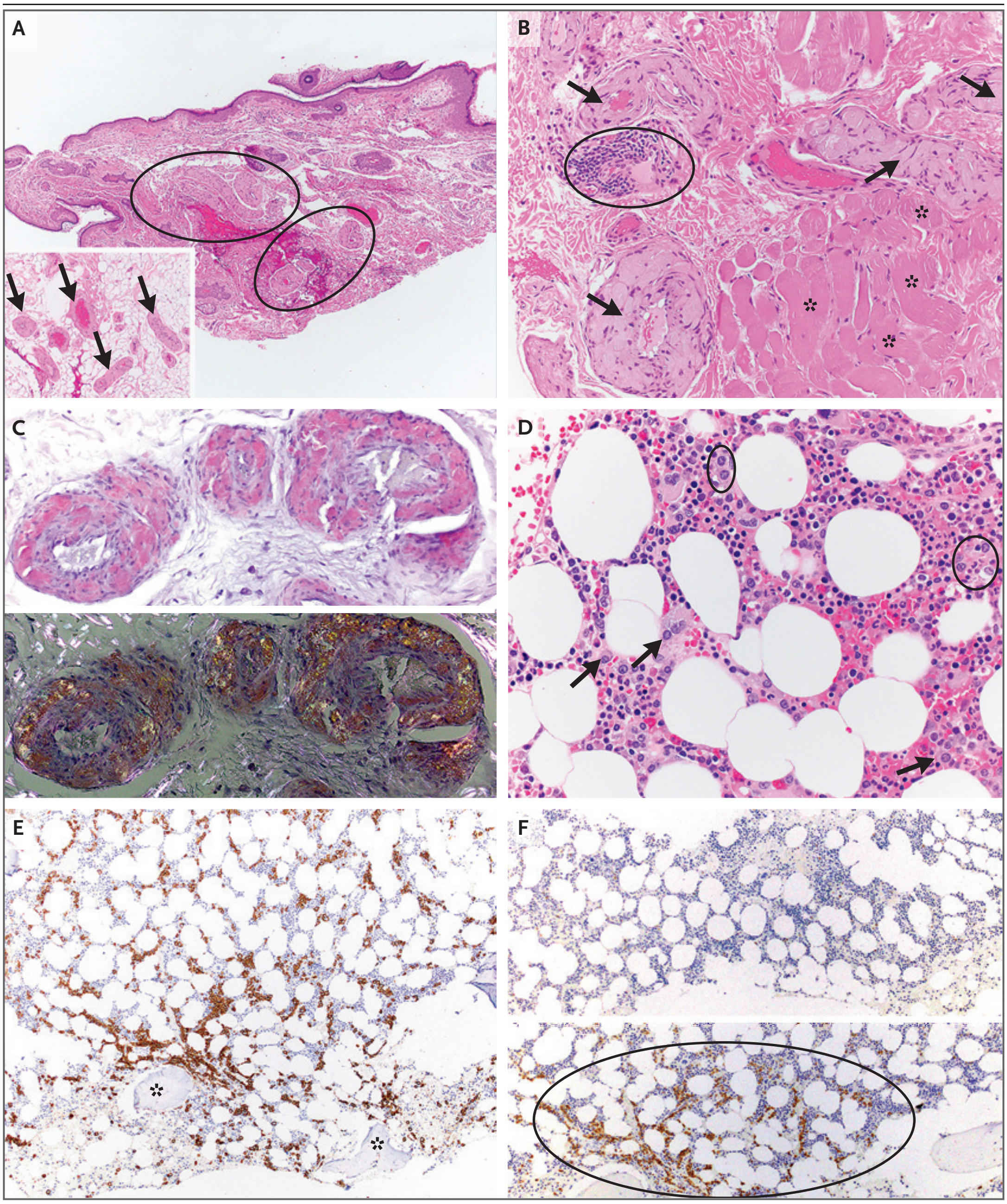
DR. NATALIE WOLKOW'S DIAGNOSIS

Systemic amyloidosis.

PATHOLOGICAL DISCUSSION

Dr. Anna M. Stagner: Hematoxylin and eosin staining of the biopsy specimen from the left upper eyelid (Fig. 3A) showed a paucity of inflammation. Vessel walls within the dermis, subcutis, and superficial skeletal muscle (orbicularis oculi) were markedly thickened and were expanded by faintly eosinophilic material. At higher magnification (Fig. 3B), the material was acellular and had a pale-pink, amorphous, homogenized appearance. On Congo red staining (Fig. 3C), the faintly eosinophilic material was highlighted in bright magenta and was associated with apple-green birefringence when viewed under cross-polarized light. Liquid chromatography with tandem mass spectrometry showed a peptide pattern consistent with lambda immunoglobulin light-chain amyloid.

After the diagnosis of amyloidosis was estab-



lished, a bone marrow biopsy (Fig. 3D) revealed evidence of a plasma-cell neoplasm, with an increased number of mature plasma cells (accounting for 15 to 20% of the marrow cellularity) in a background of normal maturing trilineage hematopoiesis. The plasma cells were highlighted on immunostaining for CD138 (Fig. 3E) and showed lambda light-chain restriction on immunostaining for kappa and lambda immunoglobulin light chains (Fig. 3F).

PATHOLOGICAL DIAGNOSIS

Deposition of light-chain amyloid in the ocular adnexa due to a systemic plasma-cell neoplasm.

DISCUSSION OF SURGICAL MANAGEMENT

Dr. Wolkow: The patient was referred to an amyloidosis center for the management of systemic

Figure 3 (facing page). Specimens from Eyelid Biopsy and Bone Marrow Biopsy.

Hematoxylin and eosin staining of the biopsy specimen from the left upper eyelid (Panel A) shows a relatively unremarkable epidermis and no substantial dermal inflammatory infiltrate. However, vessel walls within the dermis are markedly thickened and are expanded by acellular eosinophilic material (circles). The same material is seen in the vasculature of the adipose tissue in the preseptal eyelid (inset, arrows). A photomicrograph with higher magnification (Panel B) shows that the eosinophilic material has a pale-pink, amorphous appearance with a cracking artifact. The material is distributed in a strikingly perivascular pattern (arrows) and extends into the orbicularis oculi muscle (asterisks). Focal, small, admixed lymphoid aggregates are present (circle). Congo red staining (Panel C) highlights the vessel walls in bright magenta (top), and cross-polarization reveals apple-green birefringence of the congophilic material (bottom). These findings are consistent with a diagnosis of amyloidosis. Hematoxylin and eosin staining of the biopsy specimen from the bone marrow (Panel D) shows an increased number of mature plasma cells, both singly (arrows) and in small clusters (circles), in a background of maturing myeloid and erythroid elements and megakaryocytes. The plasma cells have eccentric nuclei, clumped nuclear chromatin (known as "clockface nuclei"), perinuclear clearings, and purplish-blue cytoplasm. Overall, plasma cells account for 15 to 20% of the marrow cellularity (normal range, 2 to 3%). Immunostaining for CD138 (Panel E) highlights the plasma cells (in brown); the area of highest density within the bone marrow specimen is shown, with asterisks indicating the bone spicules. Immunostaining for kappa and lambda immunoglobulin light chains (Panel F; top and bottom, respectively) shows expression of lambda light chains in the plasma cells (circle, in brown) without substantial staining for kappa light chains (normal range for kappa:lambda ratio, 2:1 to 3:1), a finding that confirms the presence of a clonal plasma-cell population.

immunoglobulin light-chain (AL) amyloidosis. However, he requested to have more upper-eyelid tissue removed because he had noticed an increase in his visual field after the eyelid biopsy.

Bilateral upper-eyelid blepharoplasty with removal of skin and fat was performed. Repair of eyelid ptosis was deferred in the context of amyloidosis with impaired levator-muscle function. Owing to amyloid deposition, the vessel walls were more friable, which resulted in increased intraoperative bleeding. After the surgical procedure, the patient noticed improvement in his vision (Fig. 4). Histopathological examination of the eyelid skin and fat excised during blepharoplasty revealed the same features that were observed in the eyelid-biopsy specimen.

Figure 4. Clinical Photographs before and after Blepharoplasty.

A photograph of the eyes obtained at the preoperative visit (Panel A) shows upper-eyelid puffiness, edema, and dermatochalasis, with skin overhanging the upper-eyelid margin. A photograph of the eyes obtained at the 3-month postoperative visit (Panel B) shows clinical improvement.

The patient has persistent diplopia, which is his most prominent source of functional impairment. He has been referred to a strabismus specialist, but the diplopia remains very challenging to treat.

DISCUSSION OF AMYLOIDOSIS MANAGEMENT

Dr. Camille V. Edwards: This patient had an underlying clonal plasma-cell disorder, as evidenced by a monoclonal spike of 1.29 g per deciliter on serum protein electrophoresis. Serum and urine immunofixation testing confirmed the presence of an IgG lambda monoclonal protein, and the serum free lambda light-chain level was 134.7 mg per liter (reference range, 5.0 to 26.3). In addition, there was bone marrow plasmacytosis, with a high burden of plasma cells in bone marrow (>10%).

Because deposition of amyloid fibrils can occur in various organs and systems, AL amyloidosis can involve the heart, kidneys, liver, peripheral nervous system (including the autonomic nervous system), gastrointestinal tract, and soft tissue.[24] In this patient, the first step in the management of his newly diagnosed AL amyloidosis was a detailed evaluation in order to assess the extent of organ involvement and select an appropriate treatment regimen.[25] His macroglossia and eye symptoms with biopsy-confirmed amyloidosis of the eyelids were indicative of soft-tissue involvement of AL amyloidosis.

An assessment for cardiac involvement included biomarker testing and imaging.[26] The B-type natriuretic peptide (BNP) level was mildly elevated (56.0 pg per milliliter; reference range, 0.0 to 53.2). The N-terminal pro-BNP level was normal (167 pg per milliliter; reference range, 0 to 899), as was the troponin I level (0.010 ng per milliliter; reference value, <0.033). Echocardiography revealed a mildly increased interventricular septal thickness (12 mm; reference value, ≤11) with focal thickening of the mid-to-upper septum (14 mm) and reduced global longitudinal strain (−15.5%). Subsequent cardiac MRI confirmed the presence of increased wall thickness but did not show late gadolinium enhancement or a radiologically significant increase in extracellular volume; the absence of these features is not consistent with the presence of cardiac amyloidosis.

The patient was also assessed for renal involvement. The estimated glomerular filtration rate, measured according to the method of the Chronic Kidney Disease Epidemiology Collaboration, was normal (>60 ml per minute per 1.73 m^2 of body-surface area). However, a 24-hour urine collection showed a protein level of 630 mg (reference range, 0 to 150). Thus, the patient was considered to have early renal involvement.[27]

Because the soft-tissue involvement was affecting the patient's vision, a treatment plan was formulated with the goal of eradicating the underlying plasma-cell clone in order to prevent further amyloid deposition that could worsen his vision. The patient had a high burden of plasma cells in bone marrow, so induction chemotherapy was indicated. He was offered standard induction therapy with six cycles of daratumumab, cyclophosphamide, bortezomib, and dexamethasone.[28] In addition, on the basis of his excellent performance status and organ function, he was considered to be a candidate for consolidation therapy with high-dose melphalan and autologous stem-cell transplantation. If the patient had an adequate (very good partial or complete) hematologic response to induction therapy, he was to receive daratumumab maintenance therapy for up to 2 years; if he did not have an adequate hematologic response to induction therapy, he was to receive high-dose melphalan in combination with autologous stem-cell transplantation.[29]

Fortunately, the patient had a very good partial hematologic response to induction chemotherapy and is now receiving daratumumab maintenance therapy. Nonetheless, he has persistent eye symptoms due to resistant amyloid fibrils that are not amenable to current plasma-cell–directed treatments.

PATIENT PERSPECTIVE

The Patient: Over a period of 16 months, I received diagnoses of glaucoma, cataracts, carpel tunnel syndrome, and heart failure. I also had a host of unexplained symptoms, such as an enlarged tongue, tingling and numbness in my feet, jaw pain when eating, and pain in the elbows, hip, and pelvis. Initially, I was very discouraged by my eyelid swelling and its effects on my vision. Fortunately, my doctor did not give up on me and instead continued to pursue the cause of my eyelid swelling.

After the biopsy of my eyelid showed amyloid, we could look back on those new diagnoses and unexplained symptoms and see the common thread of amyloidosis as the root cause. I began treatment in the amyloidosis center, and I think I am in better health than I was 2 years ago. As for my vision, it continues to worsen very slowly. The "good days" are less frequent than they were 6 months ago. When I move my eyes, it's like I'm watching an old reel-to-reel movie slowed down to the point where you can see one frame move to the next. That's what it's like — no smooth movement, just one picture after another. When I'm in the airport or some other busy location, it's dizzying and overwhelming.

One of the key things for doctors to know from my story is how important the exemplary communication among departments, facilities, and physicians across specialties was for me as a patient. Collaboration is so important with a complex disease like amyloidosis, with its ability to mimic other diseases and attack multiple body systems and its potential to cause irreversible harm over a short period of time.

FINAL DIAGNOSIS

Systemic immunoglobulin light-chain (AL) amyloidosis.

This case was presented at Ophthalmology Grand Rounds at Massachusetts Eye and Ear.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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